

Assignment 1 - Due 2/9/26 by EOD

You are hired by a retailer selling fashion clothing in the US to forecast their demand in a weekly basis. Your manager provides you with historical sales data in the file *train.csv*. The workflow is as follows:

- Every day, your model will be used to forecast the demand for the next 7 days.

Your manager gives you total freedom to create the forecasting model (could be as simple or elaborated as you wish). The only constraint that you have is that it has to be contained in a single script called *forecaster.py* with a function called *forecast_demand* (your manager provides you an example). To ensure that the script works as intended you are given a unit test script *tests.py* ensuring that the output of your model runs within the infrastructure of the company. For the same reason, you are also constrained that the script can only use packages contained in the file *requirements.txt*.

While your manager does not care about what models you use, the performance of the model will be evaluated as follows: for any time t , the predicted demand for the next 7 days is the vector $\hat{\mathbf{d}}_t = (\hat{d}_{t+1}, \dots, \hat{d}_{t+7})$, while the actual demand is the vector $\mathbf{d}_t = (d_{t+1}, \dots, d_{t+7})$. Over a test period $t = 1, \dots, T$ (just after the data you were given in *train.csv*) your model will be evaluated as¹:

$$\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{d}}_t - \mathbf{d}_t\|_2^2.$$

Task: Come up with a prediction model that performs best according to this metric.

¹Recall $\|(a, b)\|_2^2 = a^2 + b^2$.